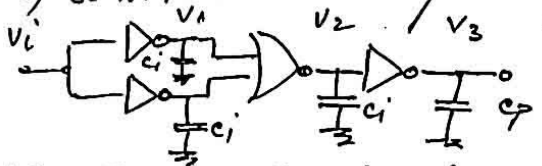
(c) 1. Poarta SI-NU - CMOS.

a) funcționare, caract. statică de transfer, margine de zgomot.

b) Să se calculeze timpul de propagare pt. o poartă fără tampon.

$C_p = 50pF$; $R_{TN} = R_{TP} = 1k\Omega$; $V_{DD} = 5^V$; $V_T = 2,5^V$

c) Să se calculeze timpul de propagare pt. o poartă cu tampon (sino).

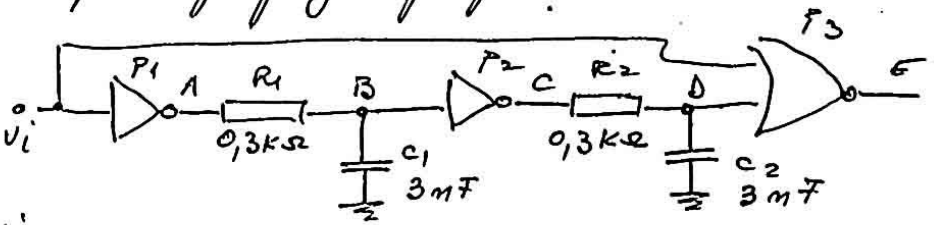


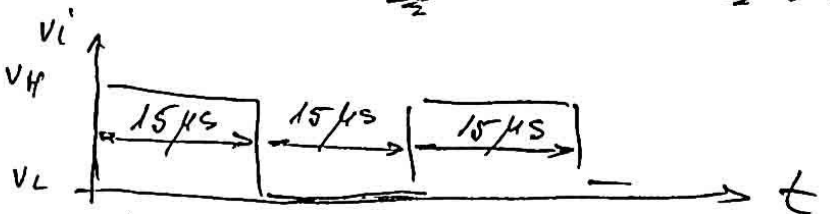
$C_p = 50pF$, $C_i = 5pF$

$R_{TN} = R_{TP} = 1k\Omega$

2. a) Se dă circ. din fig. de mai jos. Se cere să se indice diagramele de timp de funcționare în punctele A, B, C, D, E.

b) Să se determine valoarea timpilor rezultați la ieșire. Vor fi neglijați timpii de propagare pe porți.





$V_{CC} = 5^V$; $V_H = 3,5^V$; $V_L = 0,2^V$; $V_T = 1,5^V$

SUBIECT C

C1. Poarta ȘI-NU CMOS

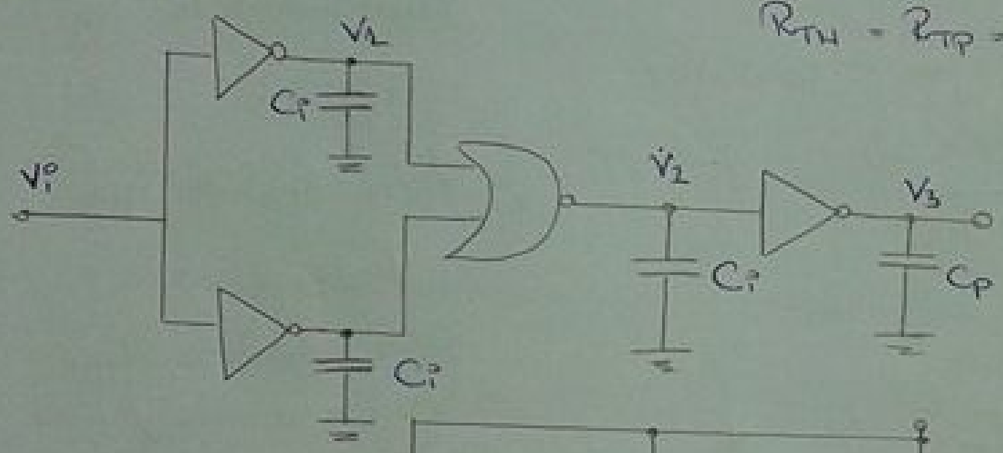
a) funcționare, caract statică de transfer, margine de zgomot

b) să se calculeze timpul de propagare pentru o poartă fără tampon

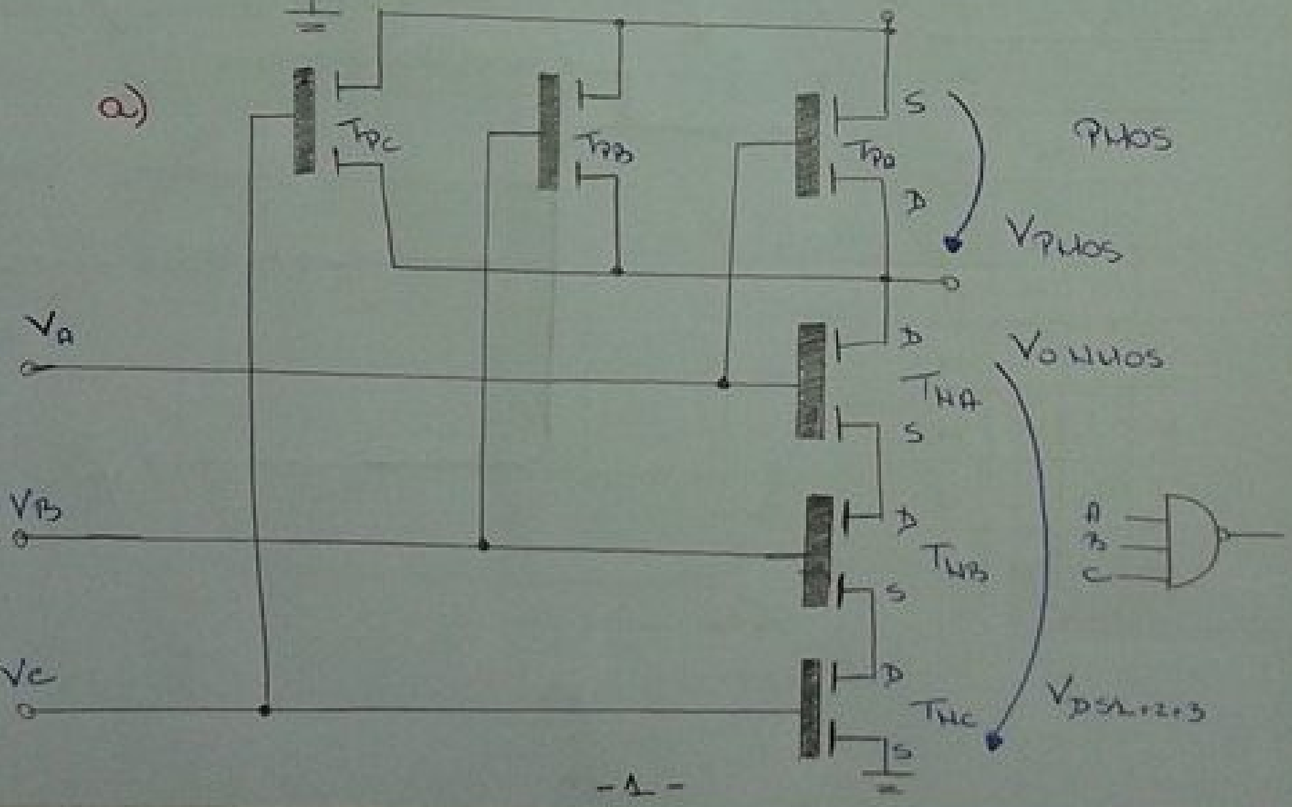
c) să se calculeze timpul de propagare pt. o poartă cu tampon (ȘI-NU)

$C_p = 50pF$; $C_i = 5pF$

$R_{TN} = R_{TP} = 1k\Omega$



a)



$V_A = V_B = V_C = V_L$ (toate tranzistoarele P conduc)

T_{PA}, T_{PB}, T_{PC} - conducție

$V_{GS} = 0 \Rightarrow T_{NC}$ - blocat $\Rightarrow$ blochează și T_{NB}, T_{NA}

$\Rightarrow V_O = V_{DD} (V_H)$

Dacă cel puțin o intrare e $V_L \Rightarrow V_O = V_{DD}$ cel puțin un T_N - blocat și T_P - conduce $\Rightarrow V_O = V_H$

b) $V_A = V_B = V_C = V_H$

T_{PA}, T_{PB}, T_{PC} - blocat

T_{NA}, T_{NB}, T_{NC} - conducție

$\Rightarrow V_O = 0V (V_L)$

$\Rightarrow F = \overline{A \cdot B \cdot C}$

V_{GS1} deschide T_{NC}

$\Rightarrow V_{D1} = 0$

V_{GS2} deschide T_{NB}

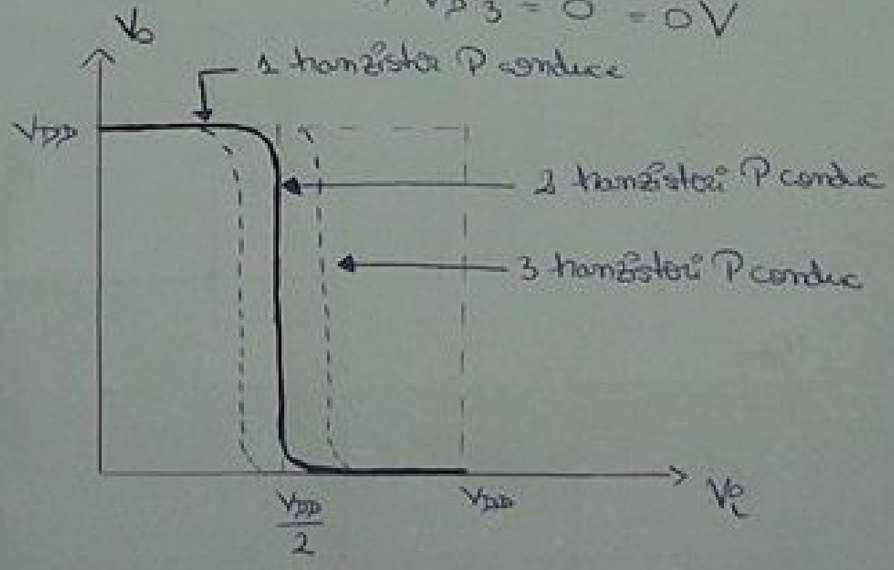
$\Rightarrow V_{D2} \simeq 0$

V_{GS3} deschide T_{NA}

$\Rightarrow V_{D3} = 0 = 0V$

• Tabel funcționare

A	B	C	F
L	L	L	H
L	L	H	H
L	H	L	H
L	H	H	H
H	L	L	H
H	L	H	H
H	H	L	H
H	H	H	L



* cu cât mai multe tranzistoare P conduc, cu atât rezistența scade, deci scade V_{PUOS} : $V_o = V_{DD} - V_{PUOS} \Rightarrow V_o$ crește

* se modifică $\tau = R \cdot C$ sau $\frac{R}{2} \cdot C$ sau $\frac{R}{3} \cdot C \Rightarrow$ mai multi timpi de propagare

• Marginea de zgomot

$M_z^+ = V_{OH\,min} - V_{IH\,min} \simeq \frac{1}{3} V_{DD}$ (ideal tinde la $\frac{V_{DD}}{2}$)

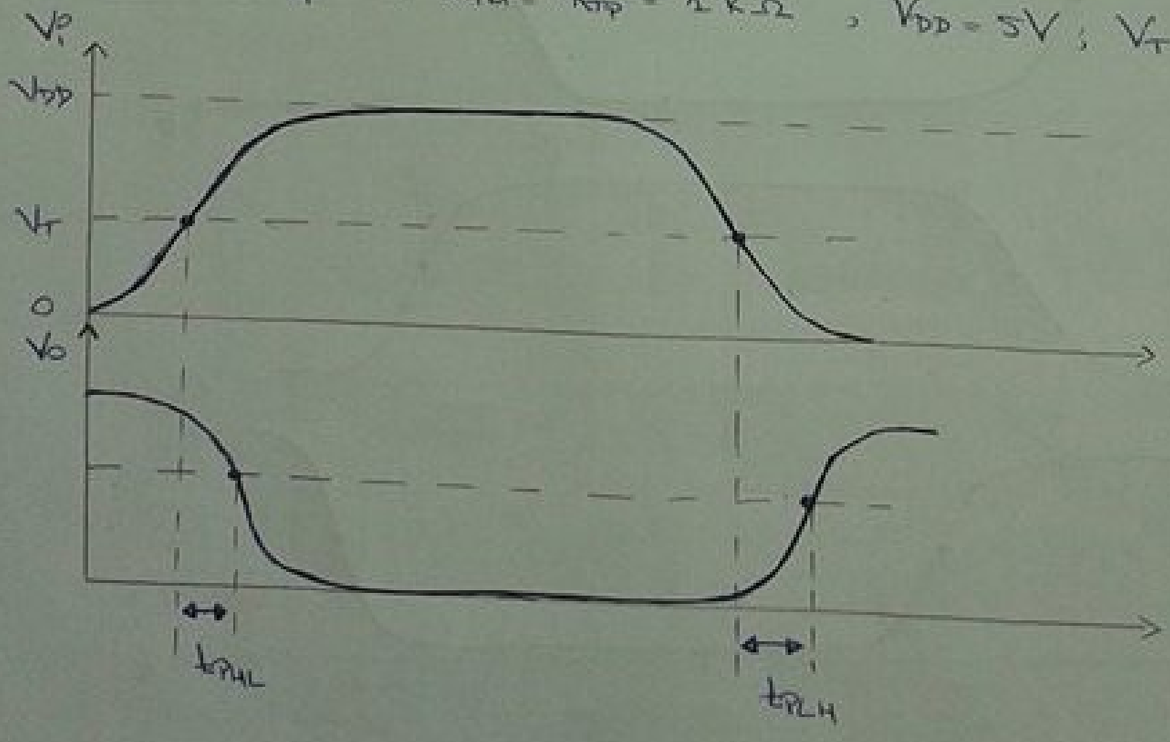
$M_z^+ = V_{DD} \left(0{,}99 - \frac{1}{1{,}5 + \frac{m_i}{m_c}}\right)$; m_i - nr. de intrări utilizate pe poartă

$M_z^- = V_{IL\,max} - V_{OL\,max} \simeq \frac{1}{3} V_{DD}$ m_c - nr. total de utilizări pe poartă

$M_z^- = V_{DD} \left(\frac{1}{1{,}5 + \frac{m_i}{m_c}} - 0{,}01\right)$

• timpul de propagare pt o poartă fără tampon

$C_p = 50\,pF$, $R_{TN} = R_{TP} = 1\,k\Omega$, $V_{DD} = 5V$; $V_T = 2{,}5V$



$$t_{PHL} = \frac{R_{TN}}{2} \cdot C_P \cdot \ln 2 = \frac{1 \cdot 10^3}{2} \cdot 50 \cdot 10^{-12} \cdot 0{,}7 =$$

$$= \frac{7}{2} \cdot 5 \cdot 10^{-9} = 17{,}5 \text{ ns}$$

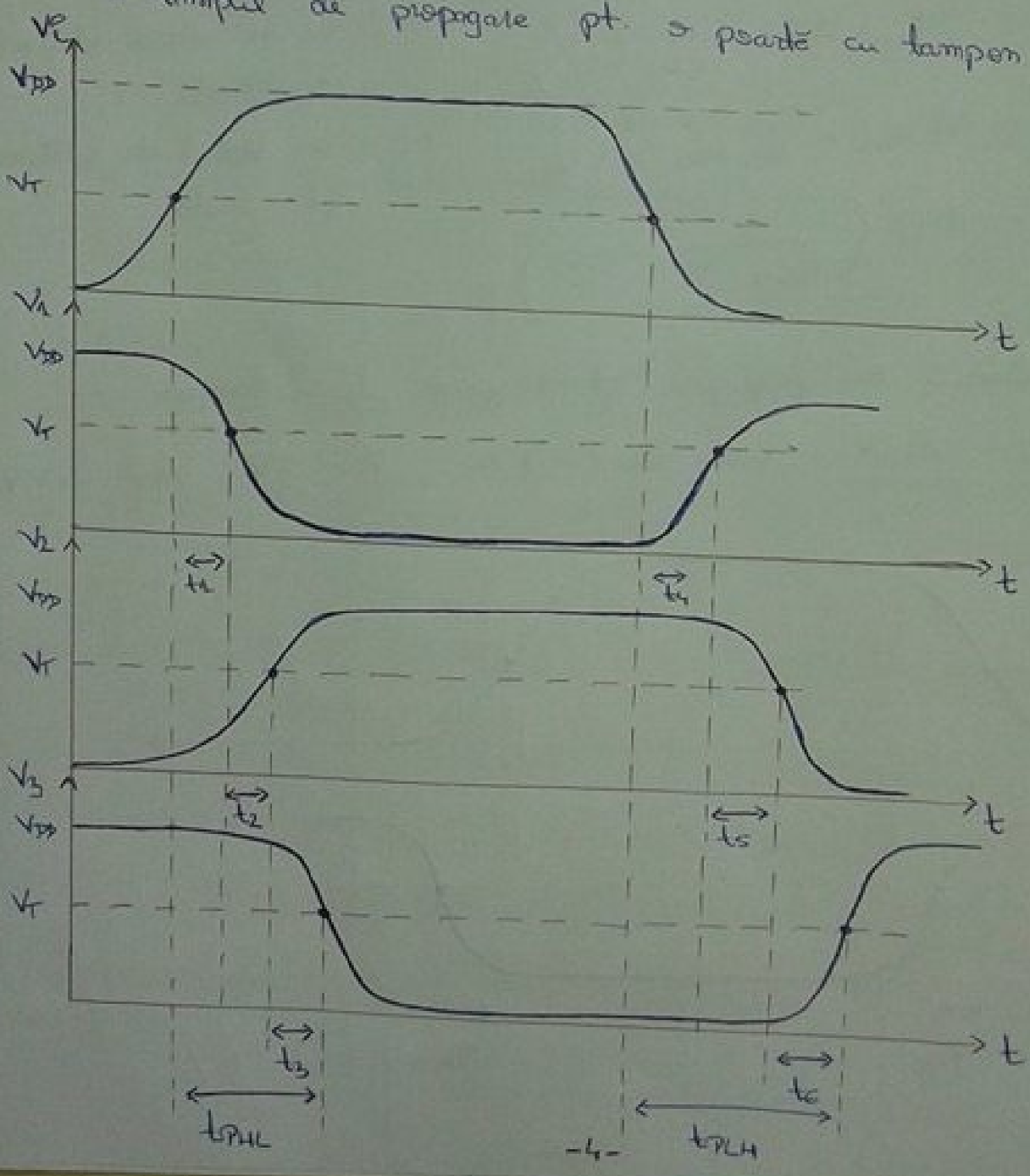
$$t_{PLH} = 2 R_{TP} \cdot C_P \cdot \ln 2 = 2 \cdot 1 \cdot 10^3 \cdot 50 \cdot 10^{-12} \cdot 0{,}7 =$$

$$= 2 \cdot 7 \cdot 5 \cdot 10^{-9} = 70 \text{ ns}$$

$$t_{PD} = \frac{t_{PHL} + t_{PLH}}{2} = \frac{17{,}5 + 70}{2} = \frac{87{,}5}{2} = 43{,}75 \text{ ns}$$

c)

- timpul de propagare pt. o poartă cu tampon



$t_{PHL} = t_1 + t_2 + t_3$

$t_1 = R_{TN} \cdot C_i \cdot \ln 2 = 1 \cdot 10^3 \cdot 5 \cdot 10^{-12} \cdot 0,7 = 0,7 \cdot 5 \cdot 10$

$t_2 = \frac{R_{TP}}{2} \cdot C_i \cdot \ln 2 = \frac{1 \cdot 10^3}{2} \cdot 5 \cdot 10^{-12} \cdot 0,7 = 1,75\ ms$

$t_3 = R_{TN} \cdot C_p \ln 2 = 1 \cdot 10 \cdot 50 \cdot 10^{-12} \cdot 0,7 = 35\ ms$

$t_{PHL} = 40,25\ ms$

$t_{PLH} = t_4 + t_5 + t_6$

$t_4 = R_{TP} \cdot C_i \cdot \ln 2 = 1 \cdot 10^3 \cdot 5 \cdot 10^{-12} \cdot 0,7 = 3,5\ ms$

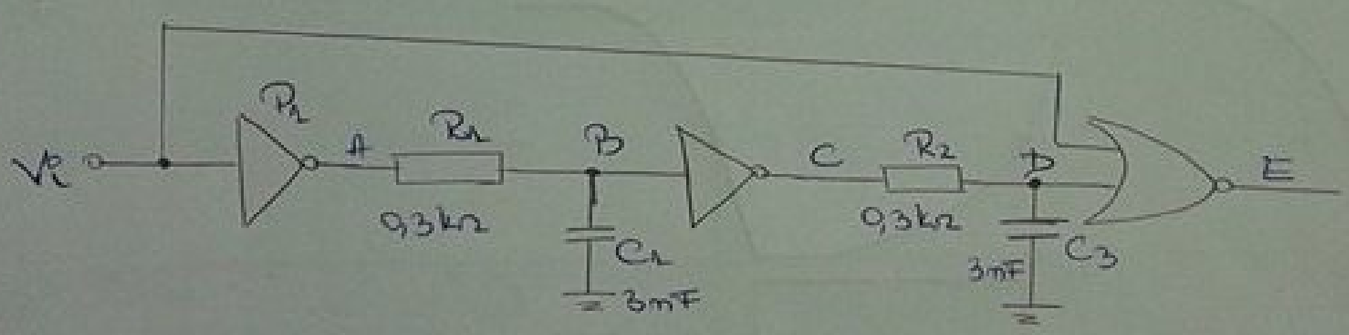
$t_5 = 2 \cdot R_{TN} \cdot C_i \cdot \ln 2 = 2 \cdot 10^3 \cdot 5 \cdot 10^{-12} \cdot 0,7 = 7\ ms$

$t_6 = R_{TP} \cdot C_p \cdot \ln 2 = 1 \cdot 10^3 \cdot 50 \cdot 10^{-12} \cdot 0,7 = 36\ ms$

$t_{PLH} = 45,5\ ms$

$t_{PD} = \frac{t_{PHL} + t_{PLH}}{2} = 42,8\ ms$

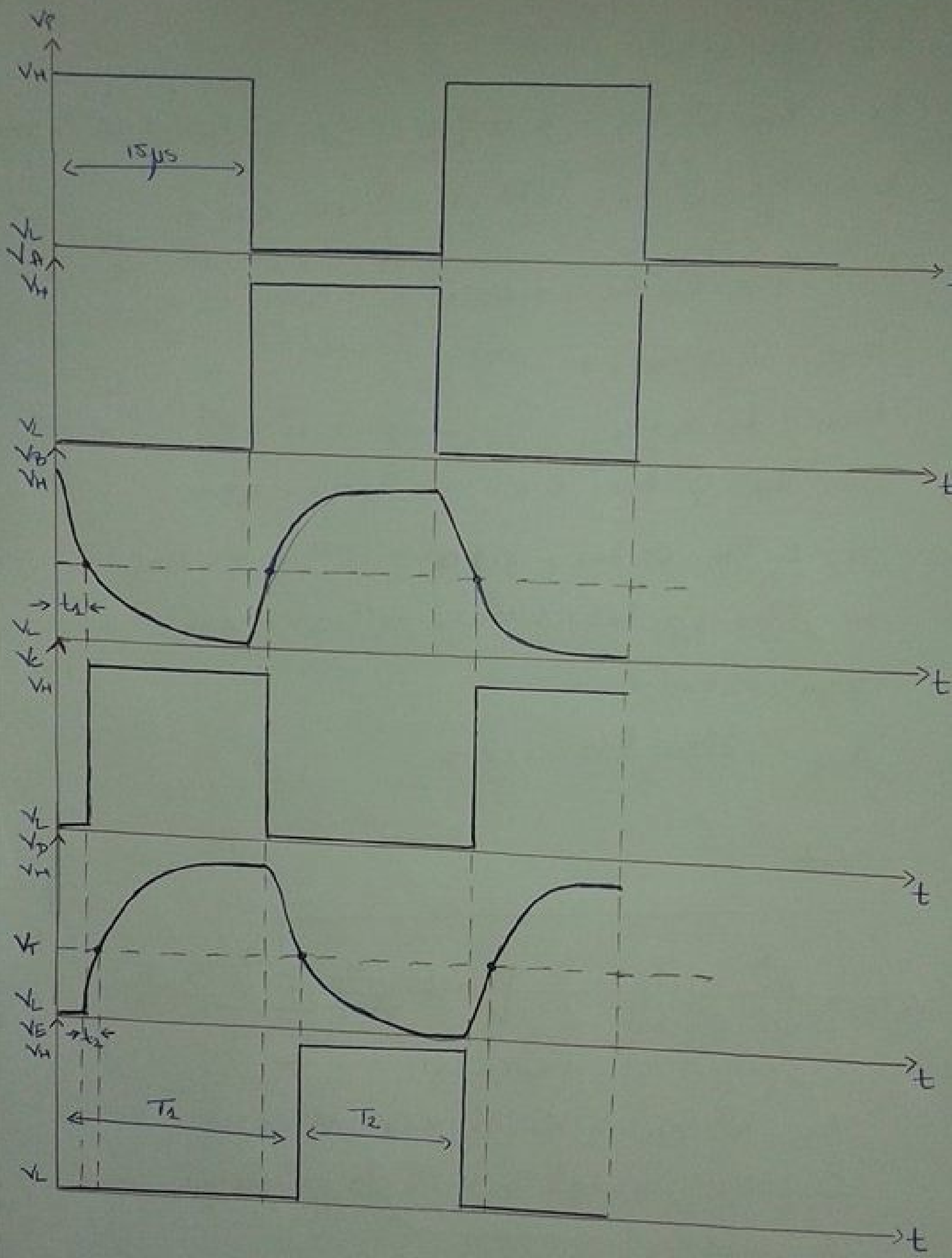
C2.



$V_{cc} = 5V$, $V_H = 3,5V$, $V_L = 0,2V$; $V_T = 1,5V$

$\tau = R \cdot C = 0,3 \cdot 10^3 \cdot 3 \cdot 10^{-9} = 0,9 \cdot 10^{-6} = 0,9\mu s < 15\mu s$

$T = 30\ \mu s$


VP
VH
15μs
VL
VA
VH
VL
VB
VH
t1
VL
VE
VH
VL
VD
VH
VT
VL
VE
VH
t2
T1
T2
VL
t

$$t_2 = R_1 \cdot C_1 \cdot \ln \frac{V_H - V_L}{V_H - V_T} = 0{,}9 \cdot \ln \frac{3{,}3}{2} = 0{,}45\,\mu s$$

$$t_1 = R_2 \cdot C_2 \cdot \ln \frac{V_L - V_H}{V_L - V_T} = 0{,}9 \cdot \ln \frac{3{,}3}{1{,}3} = 0{,}83\,\mu s$$

$$t_f = t_1 + t_2 = 1{,}28\,\mu s$$

$$\Rightarrow T_2 = 15 - (t_1 + t_2) = 15 - 1{,}28 = 14{,}75\,\mu s$$

$$T_1 = 30 - T_2 = 30 - 14{,}75 = 15{,}25\,\mu s$$